

Ghulam Ishaq Khan Institute (GIKI)

ASSIGNMENT #2

Subject: Database Management System	Course Code: CS-232-A-Spring-26
Class: BSAI – 4th, Batch: 34 (Fall – 24 Intake)	Total Marks: 10, Instructor: M. Qasim Riaz
Submission Deadline: 10/May/2026-Sun (11:59-PM)	

Project Title

Hospify — Hospital Management System

Group Members

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Project Report Component – 2 (Description)

Flow Diagram & Screenshots of the Project

Flow Diagram of the Project

The diagram below illustrates the overall user-flow of Hospify, showing how different roles navigate through the system from login to their respective modules.

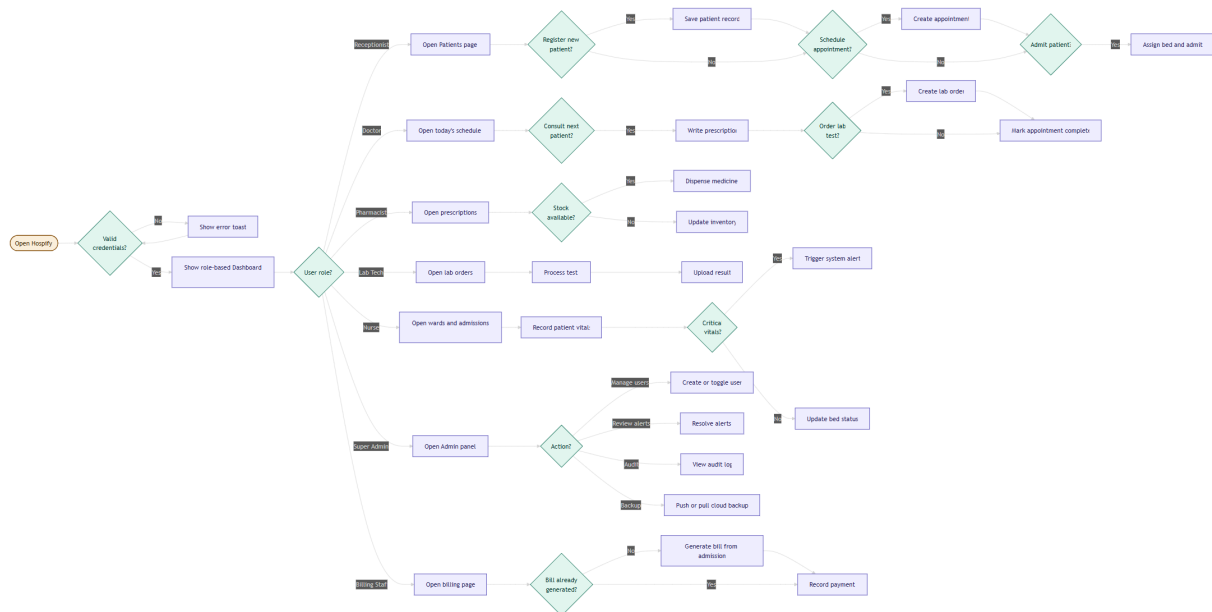


Figure 1: Hospify — System Flow Diagram

List of All Screens / Pages

1. Login Screen

Allows all system users (Super Admin, Doctor, Nurse, Receptionist, Pharmacist, Lab Technician, Billing Staff) to authenticate by entering their email and password. JWT tokens are issued upon successful login to enforce role-based access.

2. Dashboard Screen

The main landing screen after login. Displays a high-level overview of the hospital's current state including active admissions, today's appointments, bed occupancy, low-stock alerts, and recent audit activity. The view is tailored to the logged-in user's role.

3. System Admin / User Management Screen

Used by the Super Admin to create, edit, activate, or deactivate staff accounts across all roles. Displays a list of all registered users with their role and status.

4. Create User Screen

A form used by the Super Admin to register a new staff member, specifying their full name, email, password, and assigned role (Doctor, Nurse, Receptionist, etc.).

5. Patients Screen

Displays a searchable and filterable list of all registered patients. Shows key demographic details such as name, CNIC, blood group, and registration date.

6. Patient Add / Registration Screen

Used by the Receptionist to register a new patient by entering full demographic details including name, date of birth, gender, CNIC, phone, address, blood group, and emergency contact information.

7. Patient Description Screen

Shows the complete profile of a selected patient including personal details, admission history, past appointments, and linked records.

8. Search Bar Screen

A global search interface that allows staff to quickly look up patients by name or CNIC, or find records across the system.

9. Admissions Screen

Displays all current and past patient admissions. Shows patient name, assigned bed, responsible doctor, ward, admission type, and current status (active/discharged).

10. Admit Patient Screen

Used by the Receptionist to admit a patient by selecting an available bed and assigning a doctor. Triggers the `admit_patient` stored procedure which validates bed availability and logs an audit entry.

11. Appointments Screen

Shows all scheduled appointments. Allows the Receptionist to create, reschedule, or cancel appointments. Doctors can view their daily schedule and mark appointments as completed or cancelled.

12. Schedule Appointment Screen

A form used by the Receptionist to book a new appointment by selecting a patient, a doctor, a date/time, and entering the reason for the visit.

13. Doctors Screen (List)

Displays the full directory of doctors registered in the system, including their specialization, qualification, contact number, and available days.

14. Doctors Screen (Detail / Schedule)

Shows an individual doctor's daily schedule with appointment counts — pending, completed, and cancelled — powered by the `doctor_schedule_today` view.

15. Bed Availability Screen 1 (Ward Overview)

Displays a ward-level summary of bed occupancy, showing total beds, occupied, available, and maintenance counts along with the occupancy rate percentage, powered by `ward_occupancy_view`.

16. Bed Availability Screen 2 (Bed Detail)

Provides a granular view of individual beds within a selected ward, showing each bed's number, type, and real-time status (available / occupied / maintenance) synced via Firebase Firestore.

17. Pharmacy Screen

Used by the Pharmacist to view pending prescriptions from doctors and dispense medicines against prescription line items.

18. Pharmacy Stock / Inventory Screen

Displays the current medicine inventory with stock levels, reorder thresholds, and last-updated timestamps. Allows the Pharmacist to restock medicines and update quantities.

19. Lab Screen (Orders Queue)

Shows the Lab Technician's queue of pending and in-progress lab orders sorted by

priority (STAT → Urgent → Routine), powered by `pending_lab_orders_view`.

20. Lab Report / Result Upload Screen

Used by the Lab Technician to process a lab order, upload a result file, and add result summary notes to mark the order as completed.

21. Bill Screen

Used by Billing Staff to generate an itemised bill for a patient's admission, showing breakdowns for bed-days, lab tests, and prescribed medicines. Generated via the `generate_bill` stored procedure.

22. View Bill Screen

Displays the full details of a generated bill including all line items, subtotals, discounts, paid amount, and current payment status.

23. Alerts Menu Screen

Shows all active system alerts — low-stock alerts triggered by `trg_low_stock_alert` and abnormal-vitals alerts triggered by `trg_abnormal_vitals` — with options to resolve them.

24. Audit Log Screen

Accessible only to the Super Admin. Displays a full chronological log of all database changes with before/after JSONB snapshots, user responsible, action type, table name, and timestamp.

Screenshots of Each Screen

Screen 0: Login Screen (View 0)

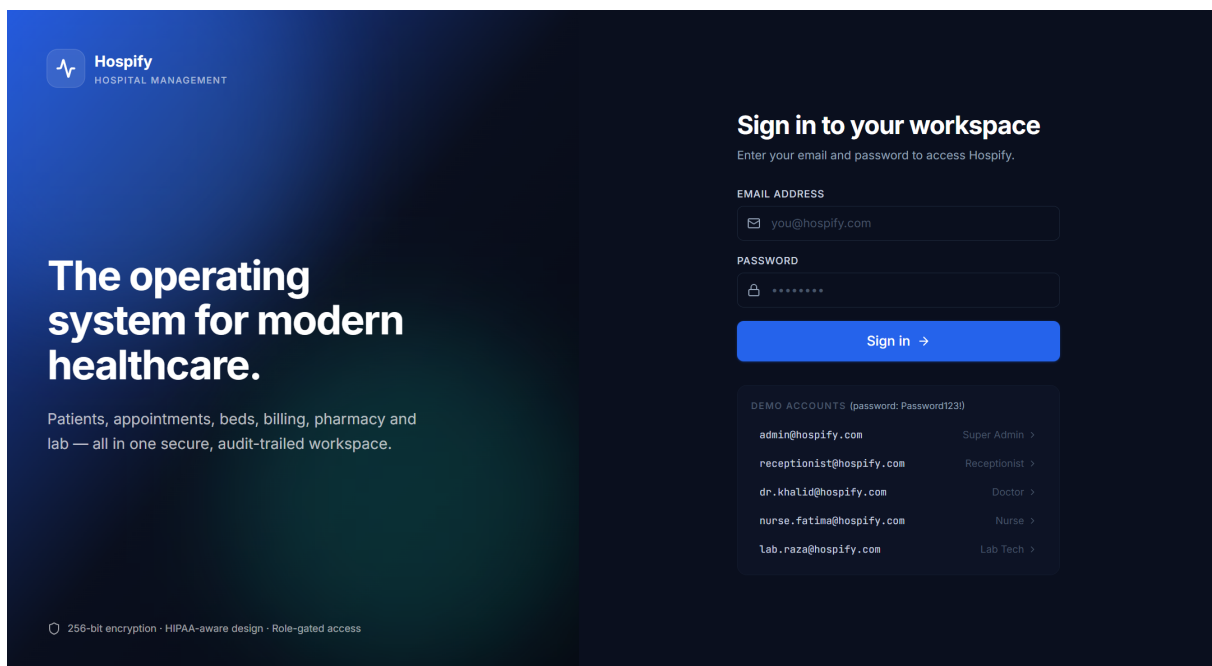


Figure 2: Screen 0: Login Screen (View 0)

Screen 1: Dashboard – Overview Panel (View 1)

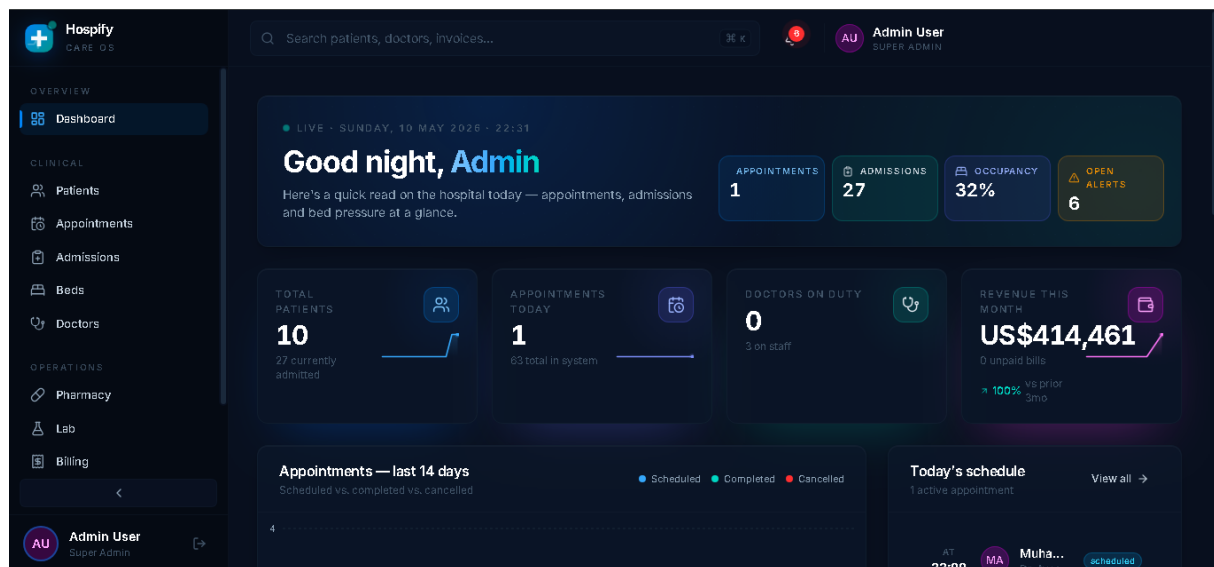


Figure 3: Screen 1: Dashboard – Overview Panel (View 1)

Screen 2: Dashboard – Overview Panel (View 2)

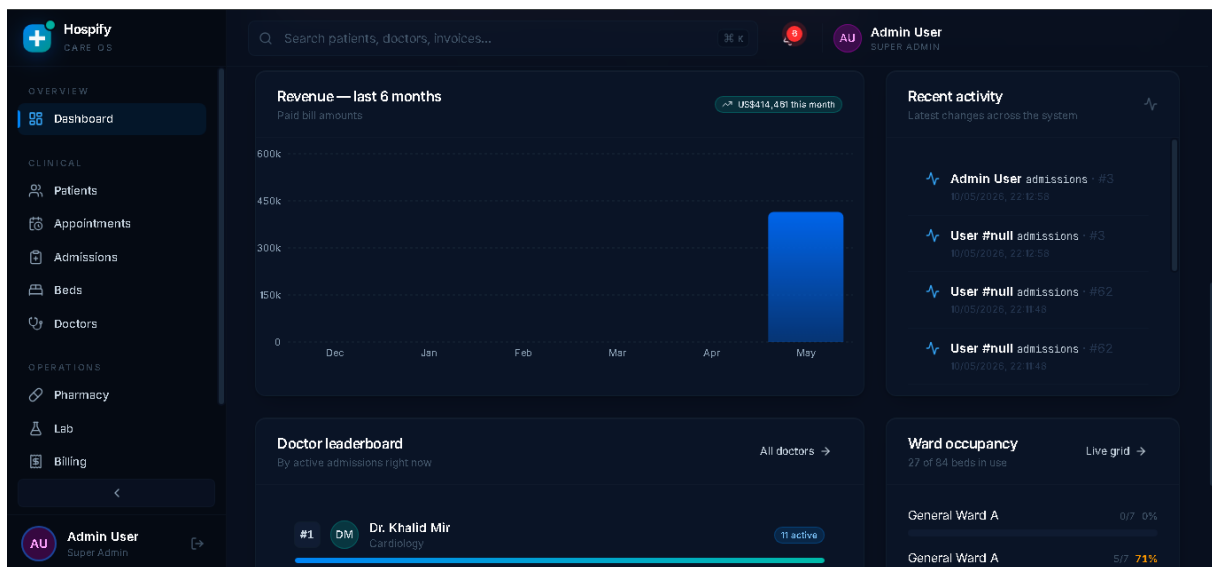


Figure 4: Screen 2: Dashboard – Overview Panel (View 2)

Screen 3: Dashboard – Overview Panel (View 3)

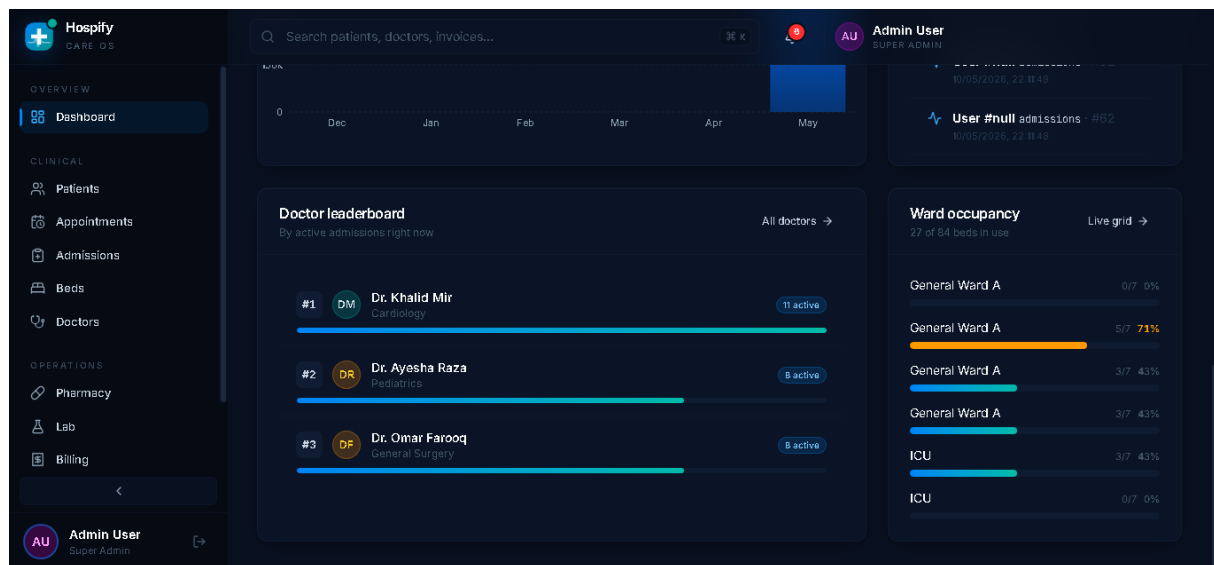


Figure 5: Screen 3: Dashboard – Overview Panel (View 3)

Screen 4: System Admin – User Management

The screenshot displays the 'System Administration' interface of the Hospify system. The left sidebar contains navigation menus for 'CLINICAL' (Patients, Appointments, Admissions, Beds, Doctors) and 'OPERATIONS' (Pharmacy, Lab, Billing). The 'SYSTEM' menu is active, showing 'Admin' as the selected option. The main content area is titled 'System Administration' and includes a search bar for patients, doctors, and invoices. Below this, there are buttons for 'Push to cloud', 'Pull from cloud', and '+ Add User'. A tabbed interface shows 'Users' and 'Audit log'. The 'Users' tab is active, displaying a table of users with columns for NAME, EMAIL, ROLE, STATUS, JOINED, and ACTIONS. The table lists four users: Tech Raza (Lab Technician), Pharma Nadia (Pharmacist), Billing Zara (Billing Staff), and Admin User (Super Admin). All users are currently 'Active' and have a 'Disable' button in the ACTIONS column.

NAME	EMAIL	ROLE	STATUS	JOINED	ACTIONS
TR Tech Raza	lab.raza@hospify.com	LAB TECHNICIAN	Active	09/05/2028	Disable
PN Pharma Nadia	pharma.nadia@hospify.com	PHARMACIST	Active	09/05/2028	Disable
BZ Billing Zara	billing.zara@hospify.com	BILLING STAFF	Active	09/05/2028	Disable
AU Admin User	admin@hospify.com	SUPER ADMIN	Active	09/05/2028	Disable

Figure 6: Screen 4: System Admin – User Management

Screen 5: Create User

Create user

×

The new user can sign in immediately with the credentials you set.

FULL NAME

EMAIL

PASSWORD

Minimum 8 characters

ROLE

doctor

▼

Cancel

+ Create

Figure 7: Screen 5: Create User

Screen 6: Patients – Patient List

The screenshot shows the 'Patients' screen in the Hospify system. The sidebar on the left contains navigation links for Overview, Clinical, and Operations. The main content area is titled 'Patients' and includes a search bar, filters, and a table of patient records.

Patient	Age	GENDER	BLOOD	Last visit	STATUS	ACTIONS
AR Ahmed Raza PT-0003 • 03211234567	55y	Male	O+	09/05/2025 1y ago	Admitted	View
AB Amna Butt PT-0006 • 03511234567	27y	Female	B-	09/05/2026 yesterday	Admitted	View
BC Bilal Chaudhry PT-0009 • 03811234567	37y	Male	A+	09/05/2025 1y ago	Admitted	View
FK Fatima Khan PT-0002 • 03111234567	33y	Female	B+	08/05/2026 2d ago	Admitted	View

Figure 8: Screen 6: Patients – Patient List

Screen 7: Patient Add / Registration

Register new patient

×

Capture the essential details — you can complete the medical history later.

FULL NAME

John Doe

DATE OF BIRTH

mm/dd/yyyy

GENDER

Male

▼

CNIC / NATIONAL ID

12345-1234567-1

PHONE

+92 ...

BLOOD GROUP

—

▼

ADDRESS

Street, City

Cancel

+ Create patient

Figure 9: Screen 7: Patient Add / Registration

Screen 8: Patient Description / Profile

Patient details

×

Full record from the medical directory.



PT-8883

Ahmed Raza

55 years

Male

 O+

 CNIC / NATIONAL ID

3520287654321

 PHONE

03211234567

 ADDRESS

Islamabad

 EMERGENCY CONTACT

Raza Mo 03212345678

 DATE OF BIRTH

05/11/1970

 REGISTERED

09/05/2026, 23:05:51

Visit history

6 records

MAY 9

Dr. Ayesha Raza
10:00 · Autism spectrum screening

scheduled

MAY 8

Dr. Ayesha Raza
14:00 · Pediatric neurology consult

completed

Figure 10: Screen 8: Patient Description / Profile

12

Screen 9: Global Search Bar

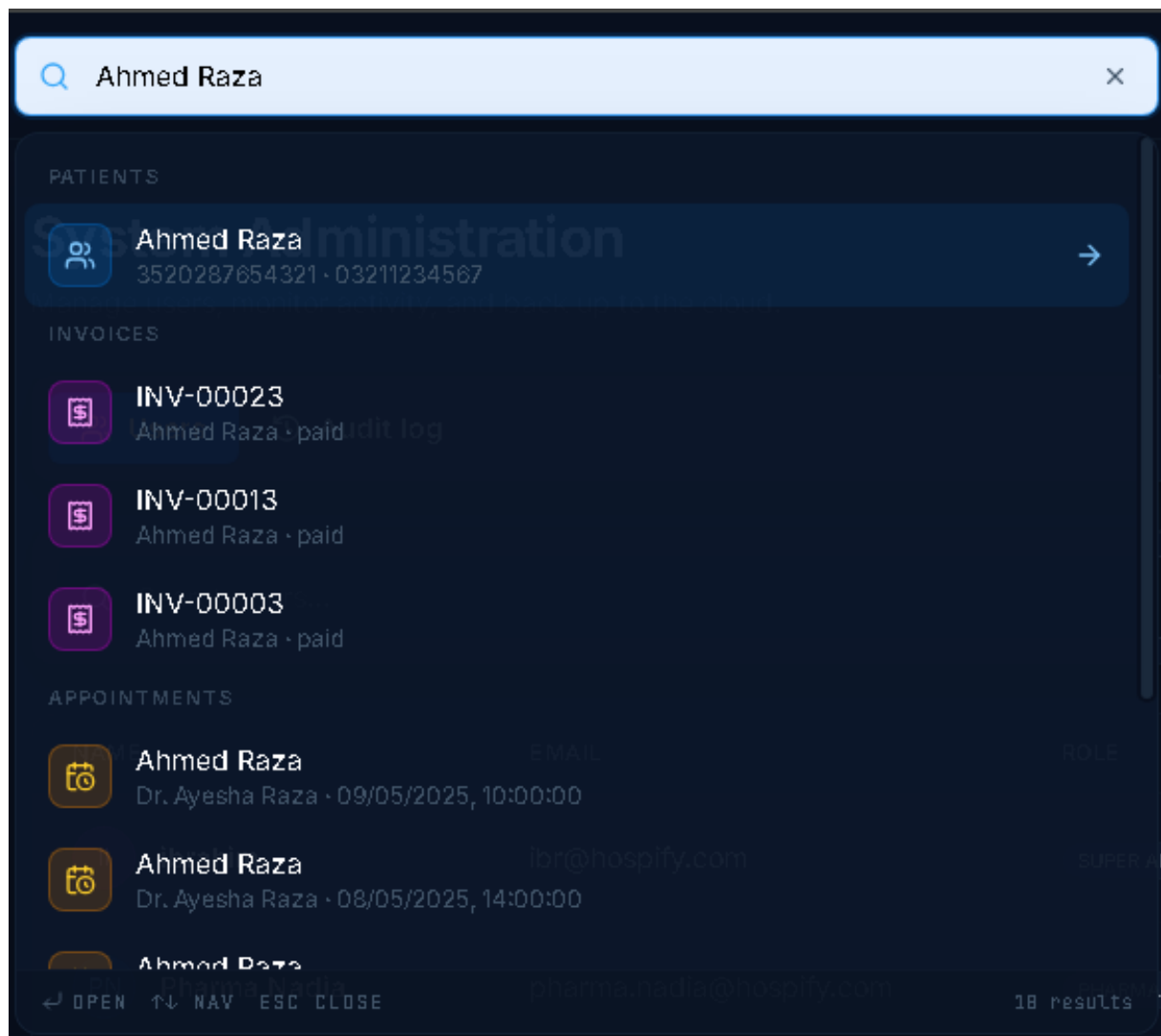


Figure 11: Screen 9: Global Search Bar

Screen 10: Admissions – Active Admissions List

The screenshot displays the 'Active Admissions' screen in the Hospify system. The sidebar on the left contains navigation links for Overview, Clinical, and Operations. The main content area shows a list of inpatients currently under care. The list includes patient details such as name, admission date, doctor, bed, and length of stay. There are buttons for 'Refresh' and 'New Admission' at the top right. The list is organized into four cards, each representing a patient's admission details.

Patient ID	Patient Name	Admission Date	Doctor	Bed	Length of Stay	Category
MA	Muhammad Ali	Admitted 01/04/2026, 16:03:47	Dr. Khalid Mir	B11-07	39d 6h	general
FK	Fatima Khan	Admitted 24/03/2026, 23:06:57	Dr. Ayesha Raza	B9-07	46d 23h	general
FK	Fatima Khan	Admitted 09/05/2026, 16:03:47	Dr. Ayesha Raza	B12-06	1d 8h	general
FK	Fatima Khan	Admitted 10/05/2026, 22:11:48	Dr. Khalid Mir	B7-02	0h	emergency

Figure 12: Screen 10: Admissions – Active Admissions List

Screen 11: Admit Patient

Admit a patient ✕

Pick a patient, a bed, and the responsible doctor.

PATIENT

Select patient ▼

ATTENDING DOCTOR

Select doctor ▼

AVAILABLE BED

Select bed ▼

ADMISSION TYPE

General ▼

Cancel + Admit

Figure 13: Screen 11: Admit Patient

Screen 12: Appointments

Hospify
CARE OS

Search patients, doctors, invoices...

Admin User
SUPER ADMIN

Appointments

Manage upcoming consultations across the hospital.

Week List Refresh + Schedule

WHEN	PATIENT	DOCTOR	REASON	STATUS	ACTIONS
MAY 10 Sun 22:00	MA Muhammad Ali 03001234567	Dr. Ayesha Raza	jkfdshikfds hkjhd	scheduled	Complete Cancel
MAY 9 Sat 18:00	AB Amna Butt 03511234567	Dr. Khalid Mir	sdgrksdfkjs	completed	
MAY 8 Fri 21:00	FK Fatima Khan 03111234567	Dr. Ayesha Raza	—	scheduled	Complete Cancel
MAY 9 Fri 12:00	AB Amna Butt 03511234567	Dr. Omar Farooq	Birth plan finalization	scheduled	Complete Cancel
MAY 9 Fri 12:00	MS Maryam Sher 03911234567	Dr. Khalid Mir	Hypertension & migraine review	scheduled	Complete Cancel
MAY 9 Fri 11:30	FK Fatima Khan 03111234567	Dr. Ayesha Raza	MRI spine follow-up	no show	

Figure 14: Screen 12: Appointments

Screen 13: Schedule Appointment

Schedule appointment

×

Pick a patient, doctor, and time.

PATIENT

Select patient

DOCTOR

Select doctor

DATE & TIME

🕒 mm/dd/yyyy --:-- -- 📅

REASON (OPTIONAL)

Brief context for the visit...

Cancel

+ Schedule

Figure 15: Screen 13: Schedule Appointment

Screen 14: Doctors – Directory List

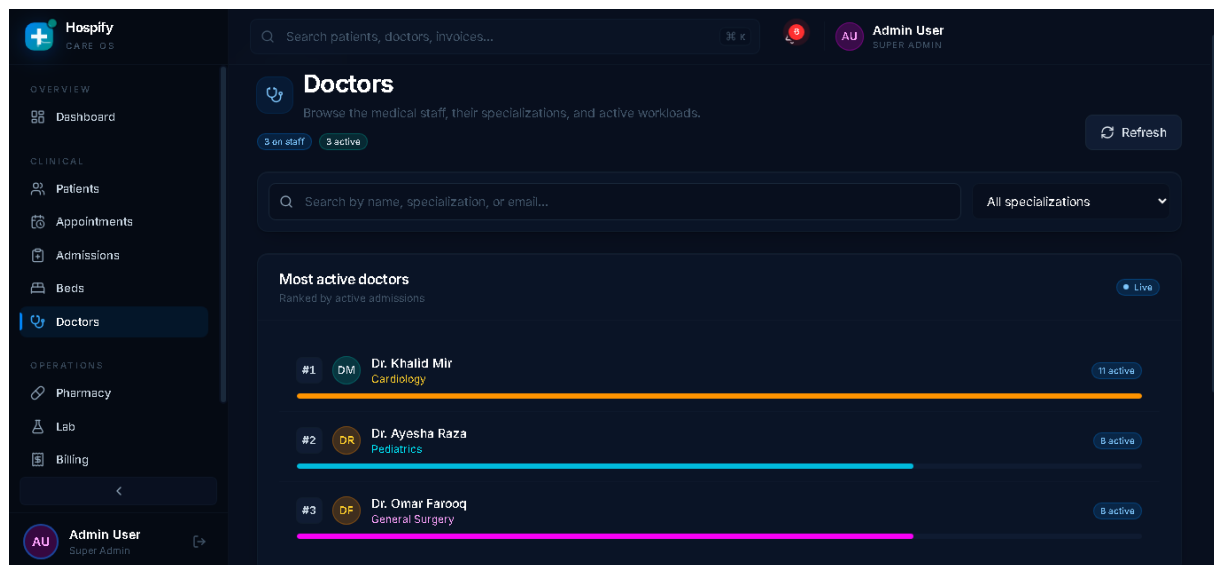


Figure 16: Screen 14: Doctors – Directory List

Screen 15: Doctors – Schedule / Detail View

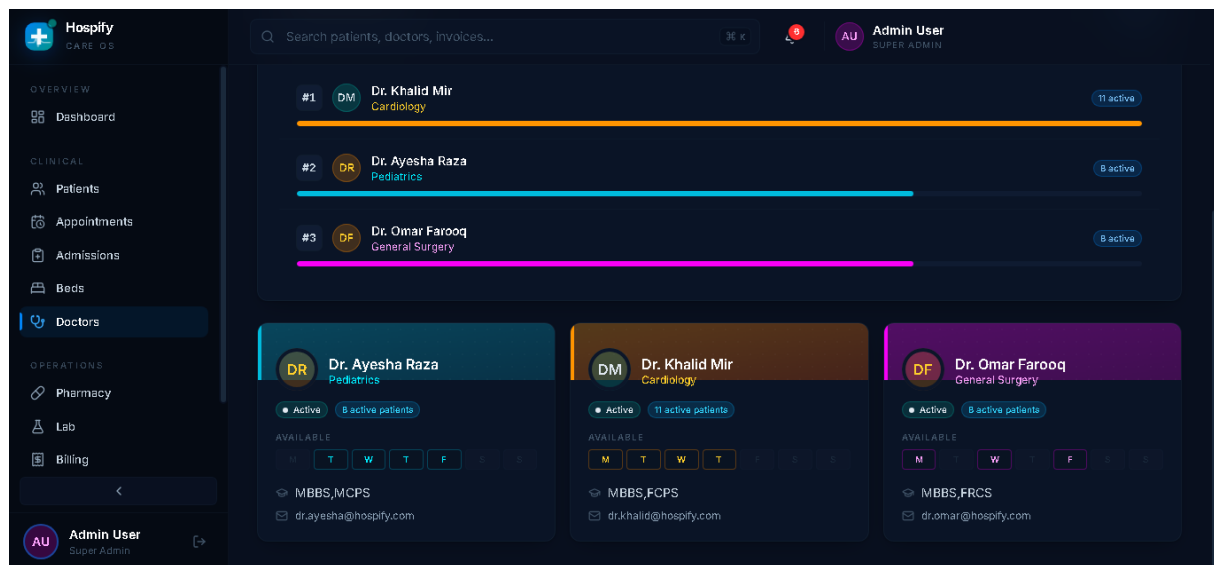


Figure 17: Screen 15: Doctors – Schedule / Detail View

Screen 16: Bed Availability – Ward Overview

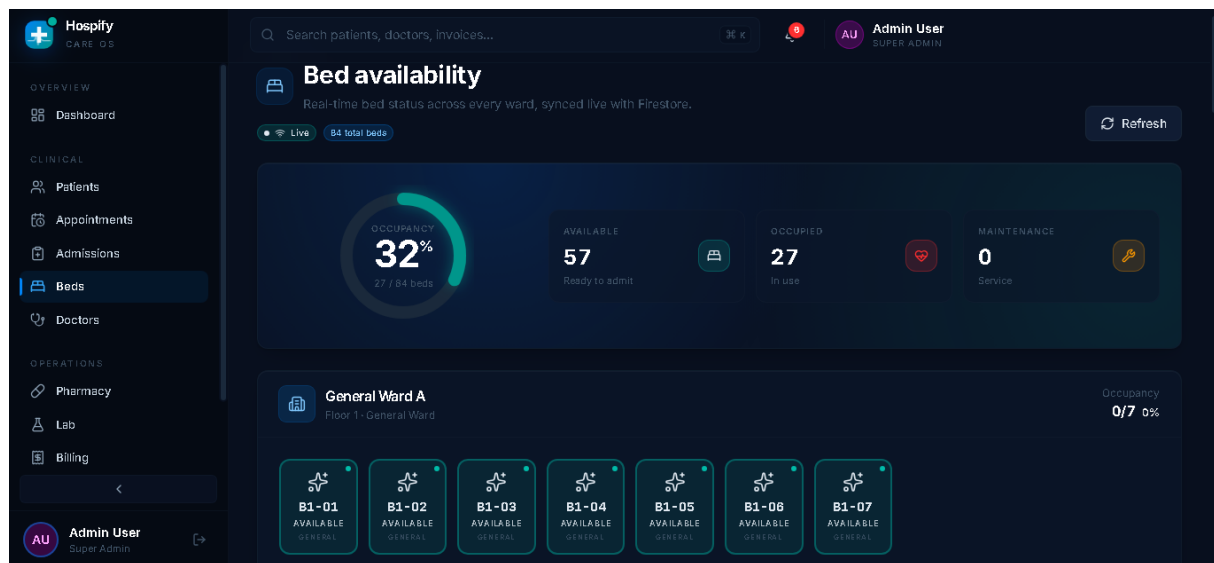


Figure 18: Screen 16: Bed Availability – Ward Overview

Screen 17: Bed Availability – Bed Detail

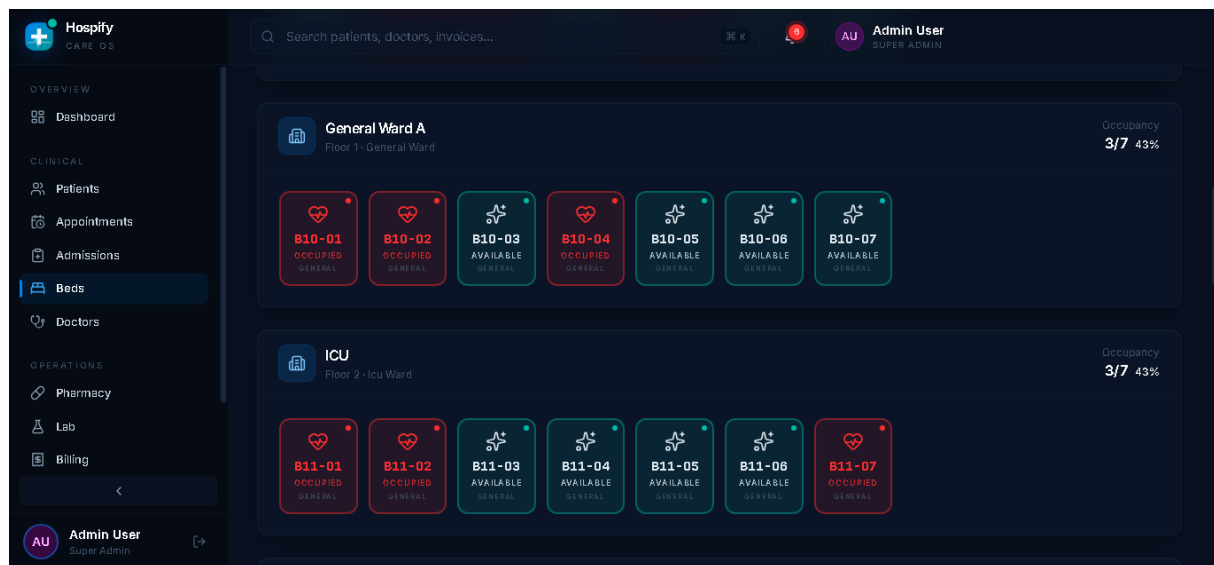


Figure 19: Screen 17: Bed Availability – Bed Detail

Screen 18: Pharmacy – Prescriptions & Dispensing

The screenshot displays the 'Pharmacy' section of the Hospify system. The interface includes a sidebar with navigation options: Overview (Dashboard), Clinical (Patients, Appointments, Admissions, Beds, Doctors), and Operations (Pharmacy, Lab, Billing). The main content area is titled 'Pharmacy' and includes a search bar for patients, doctors, and invoices. Below the search bar, there are three summary cards: 'Items in Catalog' (200), 'Total Units in Stock' (30,466), and 'Low Stock Items' (15). A table below these cards lists medicines, categories, stock levels, and status. The table has columns for Medicine, Category, Stock, Status, and Actions. Two entries for 'Elavil' (Amitriptyline - tablet) are shown, one with a stock level of 241 (In stock) and another with a stock level of 6 (Low stock). A search bar for medicines is also present, with filters for 'All', 'In stock', and 'Low'.

Pharmacy
Track medicine stock and re-order alerts.

200 Items 15 need restocking Refresh

ITEMS IN CATALOG 200

TOTAL UNITS IN STOCK 30,466

LOW STOCK ITEMS 15
At or below reorder level

Search by generic, brand, or category... All In stock Low

MEDICINE	CATEGORY	STOCK	STATUS	ACTIONS
Elavil Amitriptyline - tablet	Antidepressant	241 Reorder: 15	In stock	Update stock
Elavil Amitriptyline - tablet	Antidepressant	6 Reorder: 15	Low stock	Update stock

Figure 20: Screen 18: Pharmacy – Prescriptions & Dispensing

Screen 19: Pharmacy Stock / Inventory

Update stock

×

Elavil (Amitriptyline)

CURRENT	REORDER ≤	CATEGORY
6	15	Antidepressant

NEW QUANTITY

⬆️⬆️

⚠️ This will leave the item below the reorder threshold.

CancelSave stock

Figure 21: Screen 19: Pharmacy Stock / Inventory

Screen 20: Lab – Orders Queue

The screenshot displays the 'Laboratory Orders' section of the Hospify system. The interface includes a sidebar with navigation options: Overview, Clinical (Patients, Appointments, Admissions, Beds, Doctors), and Operations (Pharmacy, Lab, Billing). The main content area shows a list of orders with the following details:

Order ID	Test Name	Urgency	Patient Name	Doctor	Status	Action
ORDER #50	Chest X-Ray	urgent	Maryam Sher (MS) Outpatient	Ordered by Dr. Ayesha Raza	Completed	View report
ORDER #59	ECG	urgent	Bilal Chaudhry (BC) Outpatient	Ordered by Dr. Khalid Mir	Completed	View report
ORDER #58	Thyroid Panel	routine	Sana Iqbal (SI) Outpatient	Ordered by Dr. Omar Farooq	Completed	View report
ORDER #57	RFTs	stat				
ORDER #56	ECG	routine				
ORDER #55	LFTs	urgent				

Figure 22: Screen 20: Lab – Orders Queue

Screen 21: Lab Report / Result Upload

Result · Chest X-Ray×

Order #60 · Maryam Sher

TEST	RESULTED
Chest X-Ray	10/05/2026, 16:03:47
PATIENT	TECHNICIAN
Maryam Sher	Tech Raza

Summary

All values within normal reference ranges.

Figure 23: Screen 21: Lab Report / Result Upload

Screen 22: Bill Generation

Hospify
CARE OS

Search patients, doctors, invoices...

Admin User
SUPER ADMIN

FINANCE

Billing & Invoices
Generated bills, payments, and outstanding balances.

30 paid US\$0.00 outstanding Refresh

TOTAL COLLECTED
US\$414,461.00

OUTSTANDING
US\$0.00

BILLS PAID
30 of 30

Search by patient, bill ID, or admission... All statuses

INVOICE	PATIENT	GENERATED	TOTAL	PAID	STATUS	ACTIONS
INV-88838 Adm #01	Maryam Sher By Billing Zara	10/05/2026 16:03	US\$21,097.00	US\$21,097.00	paid	View
INV-88829 Adm #02	Bilal Chaudhry By Billing Zara	10/05/2026 16:03	US\$8,432.00	US\$8,432.00	paid	View

Admin User
Super Admin

Figure 24: Screen 22: Bill Generation

Screen 23: View Bill Details

INV-00030

Maryam Sher · Admission #61

TOTAL

DISCOUNT

PAID

BALANCE

US\$21,097.00

US\$0.00

US\$21,097.00

US\$0.00

Line items

SERVICE	DESCRIPTION	QTY	UNIT	TOTAL
Bed	Bed + services	3	US\$2,000.00	US\$6,000.00

Payment history

US\$21,097.00

Cash · 10/05/2026, 16:03:47

paid

Close

Figure 25: Screen 23: View Bill Details

Screen 24: Alerts Menu

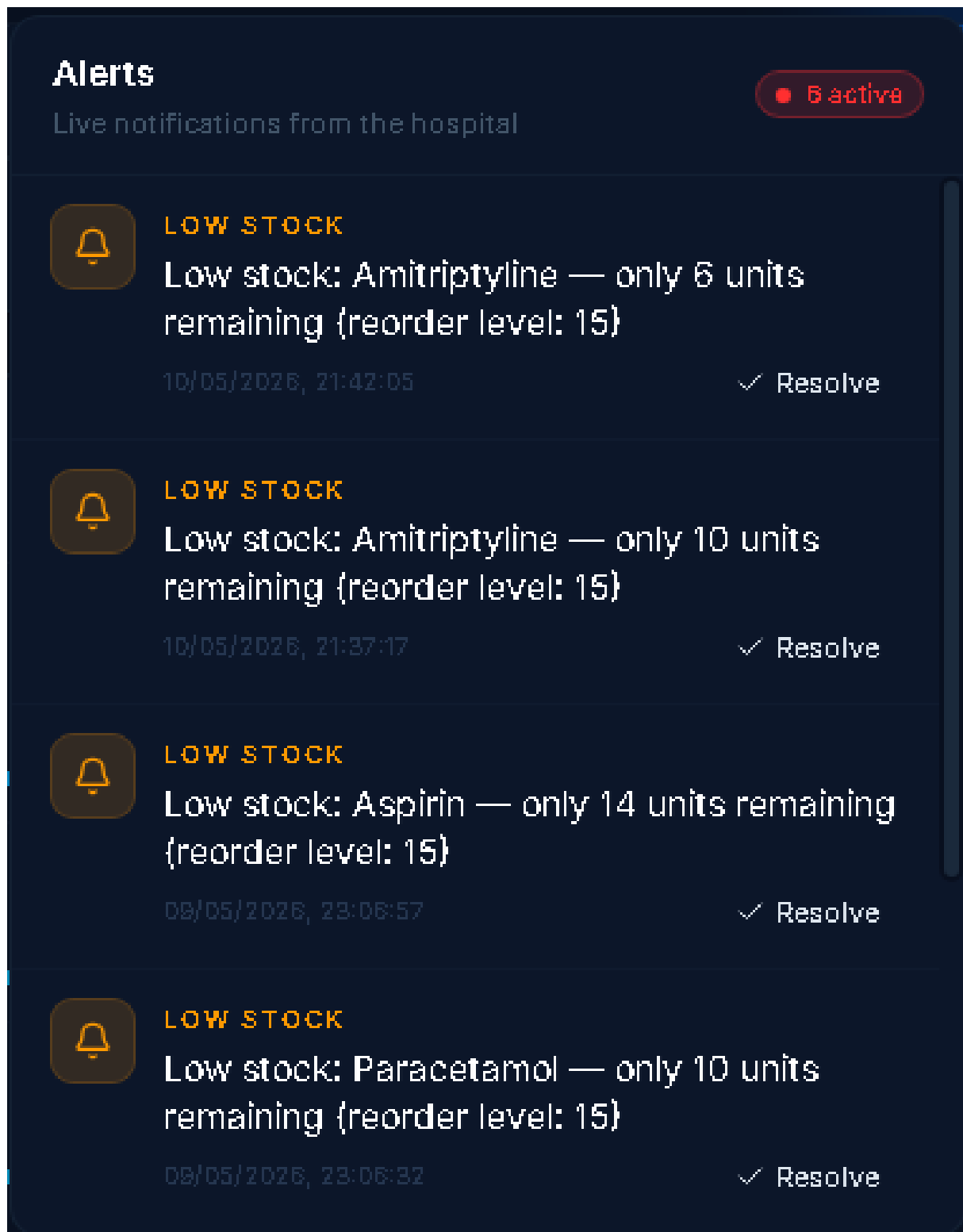


Figure 26: Screen 24: Alerts Menu

Screen 25: Audit Log

The screenshot shows the 'System Administration' interface in the Hospify CARE OS. The left sidebar contains navigation menus for 'CLINICAL' (Patients, Appointments, Admissions, Beds, Doctors) and 'OPERATIONS' (Pharmacy, Lab, Billing). The 'SYSTEM' menu is active, showing 'Admin' as the selected option. The main content area is titled 'System Administration' and includes a subtitle 'Manage users, monitor activity, and back up to the cloud.' It features buttons for 'Push to cloud', 'Pull from cloud', and '+ Add User'. Below this, there are tabs for 'Users' and 'Audit log', with 'Audit log' being the active tab. A search bar for the audit log is present, along with a 'Refresh' button. The audit log table displays the following data:

WHEN	USER	ACTION	ENTITY	CHANGES
22:12 10/05/2026	AU Admin User	—	admissions #5	—
22:12 10/05/2026	UN User #null	—	admissions #5	—
22:11 10/05/2026	UN User #null	—	admissions #62	—
22:11	UN User #null	—	admissions #62	—

Figure 27: Screen 25: Audit Log